UF Hamilton School for Classical and Civic Education
UNIVERSITY of FLORIDA

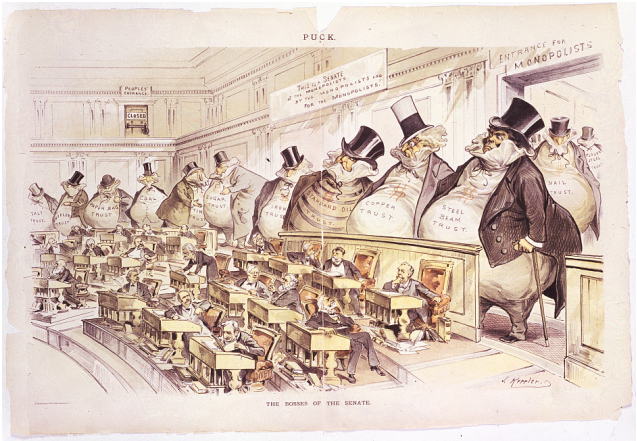
Karl Marx and His Critics

Capitalism and Its Critics
ISS 2440 — Intro to PPEL

Louis Rouanet

Fall 2026

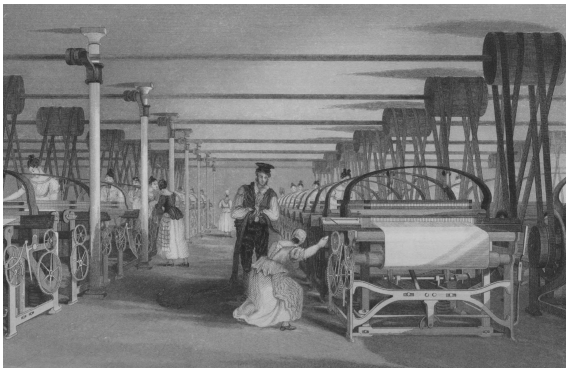
What is capitalism?



Capitalism: a working definition

- Private ownership of the means of production (land, machines, firms).
- Production for sale in markets, coordinated by prices.
- Most people work for wages, hired by the owners of capital.

Britain, 1835–1842



Housing in industrial Birmingham

Birmingham Daily Gazette, 1901

"The air is heavy with a sooty smoke and with acid vapours, and here it is that the poor live and wither away and die. [. . .] Look at the begrimed windows, the broken glass, the apertures stopped with yellow paper or filthy rags; glance in at the rooms where large families eat and sleep every day and every night, amid rags and vermin, within dank and mildewed walls from which the blistered paper is drooping, or the bit of discoloration called 'paint' is peeling away."

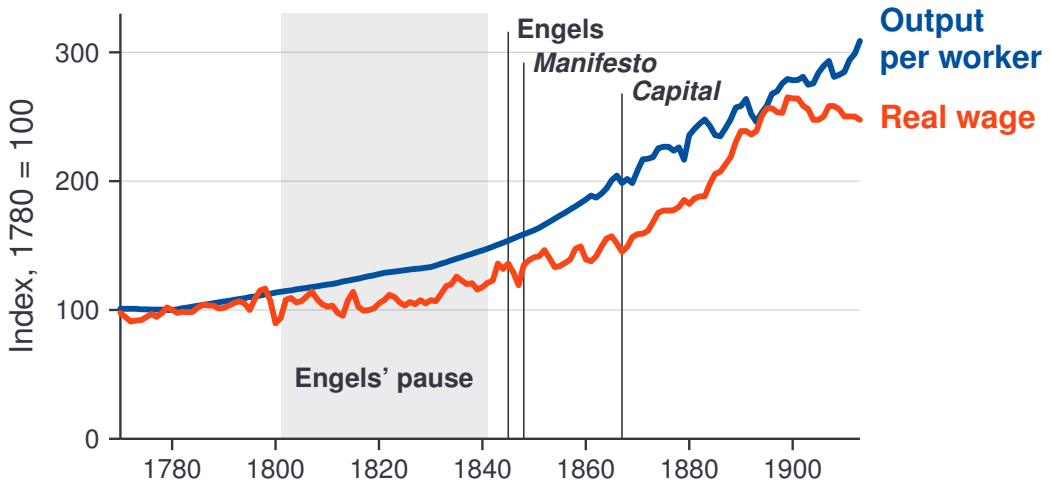
Thomas Street, Birmingham
c. 1882

Children in the coal mines

Sarah Gooder, aged 8, Children's Employment Commission (Mines), 1842

"I'm a trapper in the Gauber Pit. I have to trap without a light, and I'm scared. I go at four and sometimes half-past three in the morning, and come out at five and half-past, I never go to sleep. Sometimes I sing when I've light, but not in the dark; I dare not sing then. I don't like being in the pit."

Were workers getting better or worse off?



Note: Great Britain, output per worker and the average real wage, both 1780 = 100. Allen (2009), Fig. 1.

Engels' pause

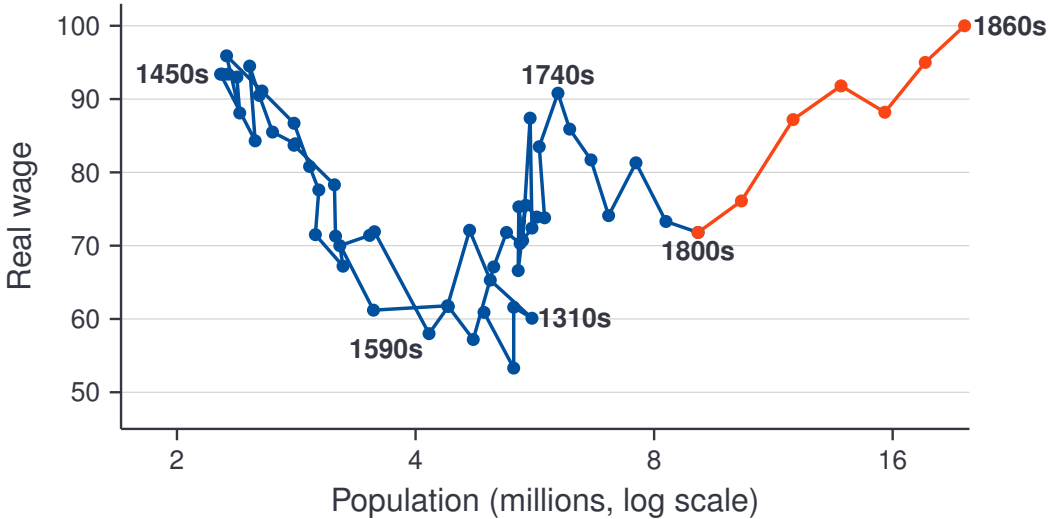
- Real wages barely grew, 1780–1830. GDP per person rose about two-thirds between 1770 and 1860.
- Productivity grew first; capital's share rose from about 20% to over 40%.
  - "Engels' pause": a half-century of delayed gains. Clearer wage catch-up came after 1840.
- Men's annual hours rose from about 2,500 to 3,300, 1760–1830: hourly earnings fared worse than annual earnings.
- Manchester grew from under 20,000 around 1750 to about 330,000 by 1850. Engels lived there in 1842–44.

Allen (2009); Voth (2001); Thomas and Dimsdale (2017); CAMPOP.

Population and wages

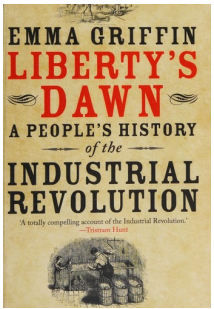
- Before 1760, population could grow about 0.4% a year without lowering real wages; after 1760, about 1.5%.
- Then earlier marriage raised births and fewer epidemics cut deaths: growth peaked at 1.5% a year around 1815.
- Births mattered more: without the rise in births, 1840 real wages would have been about 48% higher; without fewer deaths, about 27%.
- Marx and Engels saw a real shock but mistook a transient phase for a law of motion.
  - By 1900, British real wages were about twice their 1820 level.

Crafts and Mills (2022); Thomas and Dimsdale (2017).



Note: England, decade averages, 1280s–1860s. Real day wage of London building craftsmen, 1860s = 100. Allen (2001), wages; Clark (2005), population.

Liberty's Dawn: workers in their own words



- About 350 working-class autobiographies, 1760–1900, nearly all by men.
- Gains: higher and steadier earnings, literacy, political action, more choice in courtship and marriage.
- Limits: more and younger children at work; women largely shut out; the writers were literate and self-selected.

Griffin (2013).

Moving to the city: pull, push and penalty

- Pull: in the 1830s, money wages in English towns were about 73% above farm wages (Williamson).
- Push: enclosure could remove access to common land. Evidence links parliamentary enclosure to migration toward the industrial North, but not clearly to factory work.
- Penalty: life expectancy at birth was about 25–28 years in Manchester, Liverpool and Glasgow around 1840, against about 40 nationally.

Williamson (1990); Heldring et al. (2024); Davenport (2020); Davenport (2021).

Tocqueville on pauperism

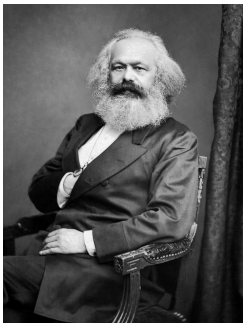
Tocqueville, *Memoir on Pauperism*, 1835

"Cross the English countryside and you will think yourself transported into the Eden of modern civilization [. . .]. Now look more closely at the villages; examine the parish registers, and you will discover with indescribable astonishment that one-sixth of the inhabitants of this flourishing kingdom live at the expense of public charity."

Alexis de Tocqueville
(1805–1859)

A note on Karl Marx, 1818–1883

- Born in Trier, 1818. Doctorate in philosophy, University of Jena, 1841.
- Edited the *Rheinische Zeitung* (1842–43) until Prussia banned it; Paris 1843–45, Brussels 1845–48.
- Wrote the *Communist Manifesto* with Friedrich Engels; published in London, February 1848.
- Lived in London from 1849 to his death in 1883, partly on Engels's money.
- *Capital*, vol. I, 1867. Engels edited vols. II and III from his drafts: 1885, 1894.

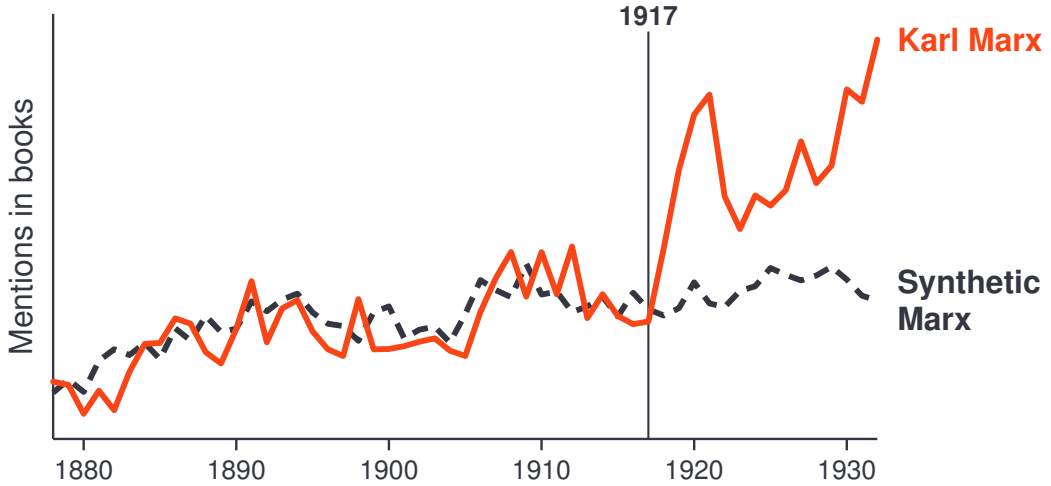


Karl Marx
(1818–1883)

Mark Blaug on Marx

Mark Blaug, *Economic Theory in Retrospect*, 1997

"Marx has been reassessed, revised, refuted and buried a thousand times but he refuses to be relegated to intellectual history."

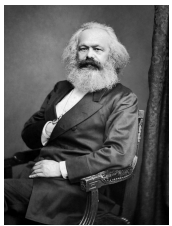


Note: How often “Karl Marx” appears in English-language books (Google Ngram), vs. a “synthetic Marx” built from other authors to track him before 1917. Magness and Makovi (2023), replication data.

Capitalism and class conflict

Marx and Engels, *Manifesto of the Communist Party*, 1848

"Our epoch, the epoch of the bourgeoisie, possesses, however, this distinct feature: it has simplified class antagonisms. Society as a whole is more and more splitting up into two great hostile camps, into two great classes directly facing each other — Bourgeoisie and Proletariat."



Karl Marx
(1818–1883)

Friedrich Engels
(1820–1895)

Utopian vs. scientific socialism

Henri de Saint-Simon
(1760–1825)

Industrialists and scientists, not nobles and priests, should run society.

Charles Fourier
(1772–1837)

Cooperative communities of about 1,600 people, housed in a “phalanstère”; work made attractive.

Robert Owen
(1771–1858)

The model mill village of New Lanark; then New Harmony, Indiana (1825–27).

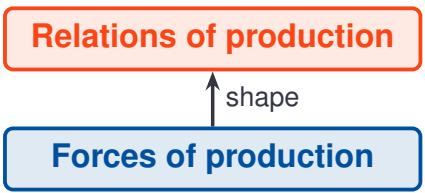
⇒ Engels: Marx’s two discoveries made socialism “a science”: historical materialism, and the secret of profit (surplus value).

Marx's core claims

- Material conditions: production, property and class drive historical development and shape ideas and consciousness.
- Dialectics: contradictions, especially class conflicts, drive social change.
- Human beings are social and productive. Labor is a means of self-realization.
- Under capitalism, workers are alienated from their product, their activity, fellow humans and their species-being.

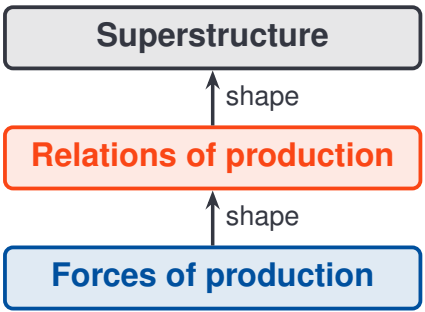
Marx (1844); Marx (1859); Marx and Engels (1848).

Forces of production Tools, machines, skills



Who owns what:
lord and serf, capitalist and worker

Tools, machines, skills

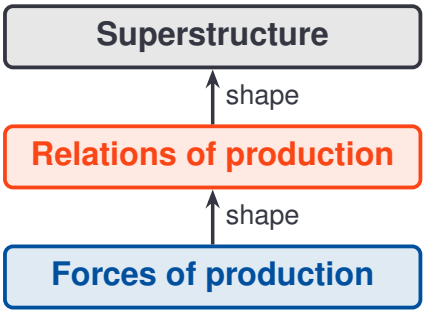


Law, politics, religion, ideas

Who owns what:
lord and serf, capitalist and worker

Tools, machines, skills

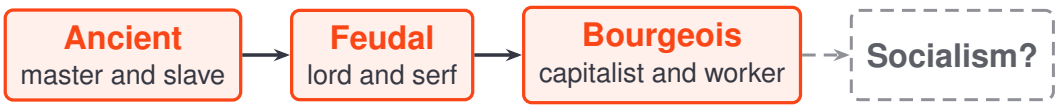
Historical materialism



Law, politics, religion, ideas

Who owns what:
lord and serf, capitalist and worker

Tools, machines, skills



Ploughs, hand tools

Hand-mills, crafts

Steam power, machinery

The hand-mill and the steam-mill

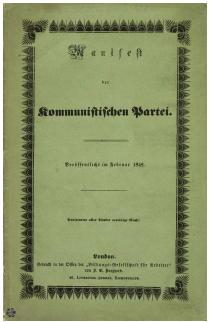
Marx, *The Poverty of Philosophy*, 1847

"The hand-mill gives you society with the feudal lord; the steam-mill, society with the industrial capitalist."

Capitalism's productive power

Marx and Engels, *Manifesto of the Communist Party*, 1848

"The bourgeoisie, during its rule of scarce one hundred years, has created more massive and more colossal productive forces than have all preceding generations together."

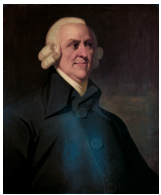

Manifest
der
Kommunistischen Partei.
London.

Manifest der Kommunistischen Partei
London, February 1848

Smith, Ricardo and Marx on value

- The classical questions: what sets relative prices, and how output splits into wages, profits and rents.
- Smith: labor alone sets exchange values only in the "early and rude state" before capital and land are owned: one beaver for two deer.
- Ricardo: relative labor quantities regulate relative values, but goods that "cannot be brought to market in the same time" differ in value.

$\Rightarrow$ Marx: labor is the substance of value, so profit too must be explained by labor.



Adam Smith
(1723–1790)

David Ricardo
(1772–1823)

Labor vs. labor power

- Labor is the work; labor power is the capacity to work.
  - The worker sells labor power by the day.
  - A mill owner who hires a spinner pays for his capacity to work that day, not for the yarn he spins.
- Marx, *Wage Labour and Capital*, ch. 2: “Labour-power, then, is a commodity, no more, no less so than is the sugar.”

Labor vs. labor power

- Labor is the work; labor power is the capacity to work.
  - The worker sells labor power by the day.
  - A mill owner who hires a spinner pays for his capacity to work that day, not for the yarn he spins.
- Marx, *Wage Labour and Capital*, ch. 2: “Labour-power, then, is a commodity, no more, no less so than is the sugar.”

If the source of exchange value is labor,
then where does profit come from?

Marx’s Theory of Capitalism

What do a coat and 20 yards of linen have in common?

Socially necessary labor time

- Marx: two goods that trade as equals “must therefore be equal to a third, which in itself is neither the one nor the other.”

Socially necessary labor time

- Marx: two goods that trade as equals “must therefore be equal to a third, which in itself is neither the one nor the other.”
- The third thing is labor. Value is the labor time needed “under the normal conditions of production, and with the average degree of skill and intensity prevalent at the time.”
- A slow tailor’s coat is not worth more: only socially necessary time counts.

Marx (1867), vol. I, ch. 1.

Selling to buy vs. buying to sell

Commodity (C): a good made to be sold. M: money.

Selling in order to buy

- The aim is use value, not profit.
- No surplus value is created. Value is conserved in exchange.

Selling to buy vs. buying to sell

Commodity (C): a good made to be sold. M: money.

Selling in order to buy

Buying in order to sell

- The aim is use value, not profit.
- No surplus value is created. Value is conserved in exchange.
- The aim is $M' > M$. The cycle is profit-driven, not consumption-driven.

Selling to buy vs. buying to sell

Commodity (C): a good made to be sold. M: money.

Selling in order to buy

Buying in order to sell

- The aim is use value, not profit.
- No surplus value is created. Value is conserved in exchange.
- The aim is $M' > M$. The cycle is profit-driven, not consumption-driven.

Where does the extra come from?

The source of surplus value

- In exchanges between commodities of equal value, no surplus value emerges.
- Yet aggregate profit is positive.
- Marx's answer $\Rightarrow$ one commodity is special: labor power. Its use can produce more value than it costs to buy.

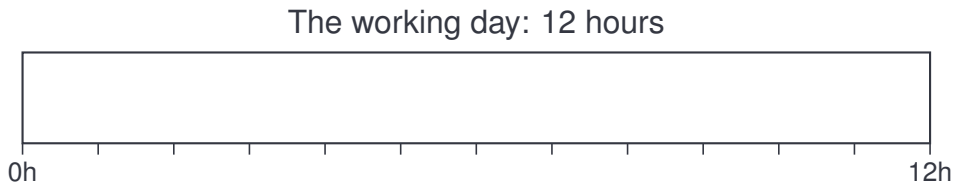
Marx (1867), vol. I, chs. 5–6.

Labor power

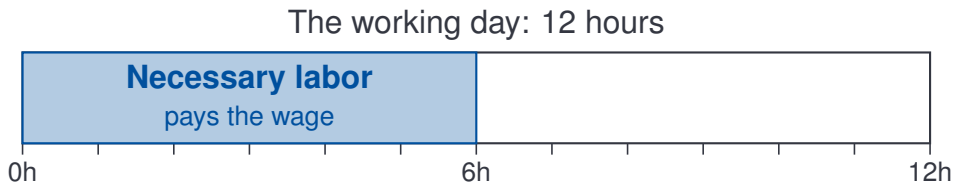
Labor power: the capacity to work, which the worker sells to the capitalist by the day.

- Like any other commodity, the value of *labor power* is the labor time socially necessary to produce and reproduce it.
- Its value = the labor time needed to produce the worker's means of subsistence, the goods he lives on: in Marx's example, **6** hours a day.
- Marx, *Capital* I: what counts as subsistence has "a historical and moral element" $\Rightarrow$ not a bare biological minimum.

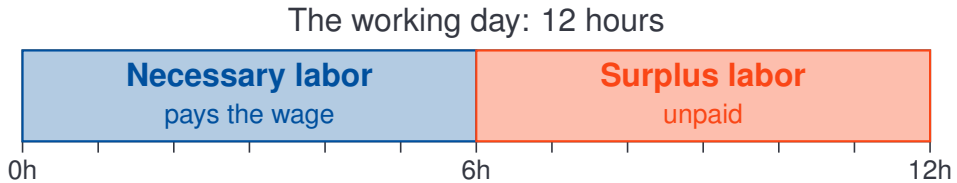
The working day



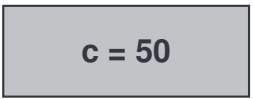
The working day



The working day



Constant capital, variable capital and surplus value



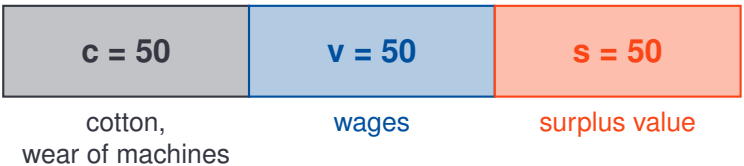
cotton,
wear of machines

- **c**, constant capital: machines and materials pass on their value, unchanged.

Constant capital, variable capital and surplus value

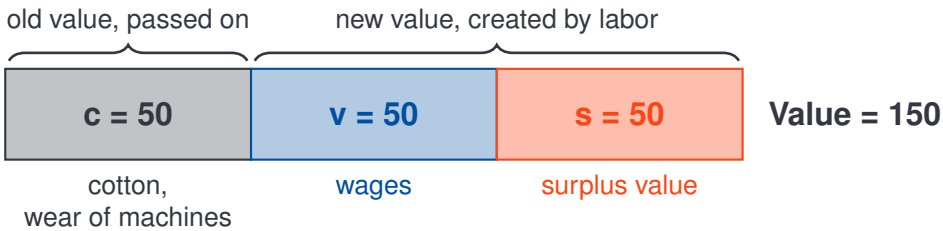
- **c**, constant capital: machines and materials pass on their value, unchanged.
- **v**, variable capital: labor power, says Marx in *Capital* I, “reproduces the equivalent of its own value, and also produces an excess”.

Constant capital, variable capital and surplus value



- **c**, constant capital: machines and materials pass on their value, unchanged.
- **v**, variable capital: labor power, says Marx in *Capital* I, “reproduces the equivalent of its own value, and also produces an excess”.
- **s**, surplus value: that excess, the value made in the **6** unpaid hours.

Constant capital, variable capital and surplus value



- **c**, constant capital: machines and materials pass on their value, unchanged.
- **v**, variable capital: labor power, says Marx in *Capital* I, “reproduces the equivalent of its own value, and also produces an excess”.
- **s**, surplus value: that excess, the value made in the **6** unpaid hours.

Exploitation, a technical term

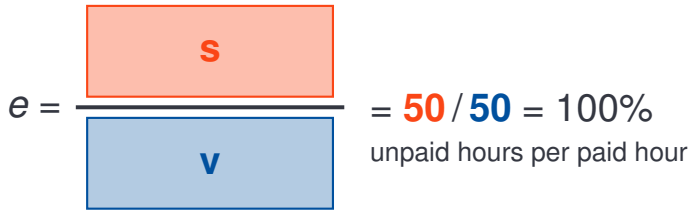
Rate of surplus value (e): unpaid hours per paid hour, Marx's measure of exploitation.

$$e = \frac{s}{v} = 50/50 = 100\%$$

unpaid hours per paid hour

Exploitation, a technical term

Rate of surplus value (e): unpaid hours per paid hour, Marx's measure of exploitation.

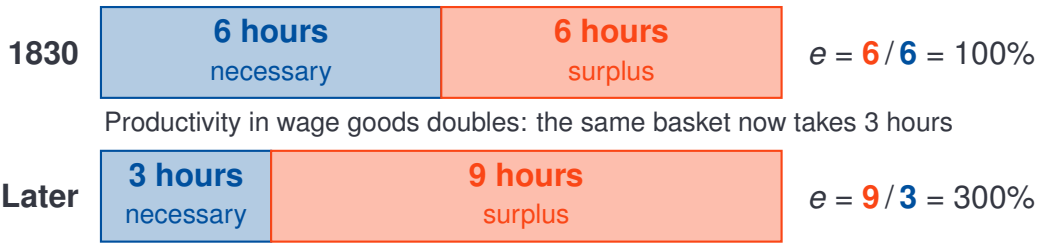


- Mechanism: **6** unpaid hours, yet the wage is the full value of labor power. No fraud.
- Moral charge: Marx calls e "an exact expression for the degree of exploitation of labour-power by capital".
- Testable claim: what Marx predicts for wages, profits and hours as capital grows.

Marx (1867), vol. I, chs. 7–9.

Relative surplus value

Relative surplus value



- Relative surplus value: the necessary part of the same day shrinks from **6** to **3** hours.
- *e* rises from 100% to 300%, while the worker buys the same basket.
- Marx: the price of labor power can keep falling while “the mass of the labourer’s means of subsistence” grows.

Marx (1867), vol. I, chs. 12 and 17.

Dead labor vs. living labor

Marx, *Capital*, vol. I, 1867

"Capital is dead labour, that, vampire-like, only lives by sucking living labour, and lives the more, the more labour it sucks."

Nosferatu, 1922

Free in a double sense

Means of production: the land, tools, machines and materials that work needs.

- The worker is “free in the double sense”: free to sell his labor power, and free of any means of production.
- Marx: capitalism began with “the historical process of divorcing the producer from the means of production”.
- Marx’s name for it: primitive accumulation. Not Smith’s story of saving and specialization.

Marx (1867), vol. I, chs. 6 and 26.

Primitive accumulation

Marx, *Capital*, vol. I, 1867

"And the history of this, their expropriation, is written in the annals of mankind in letters of blood and fire."

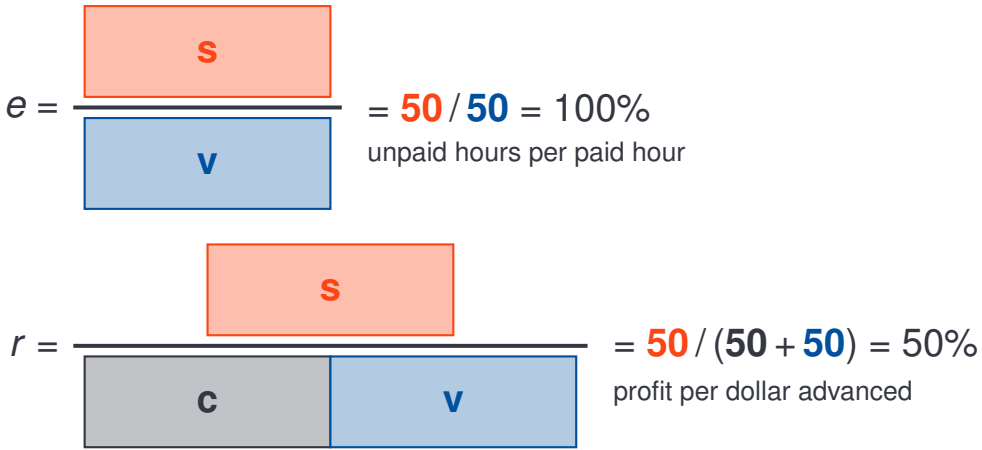
Thomas Faed, *The Last of the Clan*, 1865

The rate of profit

$$e = \frac{s}{v} = 50/50 = 100\%$$

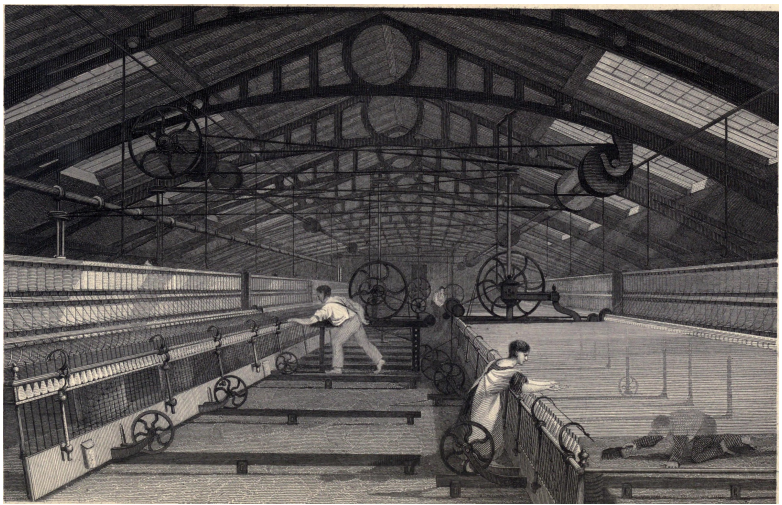
unpaid hours per paid hour

The rate of profit



Same mill, two ratios: **s** against wages (exploitation), **s** against all capital (profit).

Why buy machines?



Machines and extra profit

- The first mill to mechanize makes cloth in less than the socially necessary time and sells near the old value: an “extra surplus-value”.
- Rivals copy. The extra “vanishes, so soon as the new method of production has become general”.
- The mill now has more **c** per worker.

Marx (1867), vol. I, ch. 12.

$$r = \frac{s}{c \quad v}$$

surplus stays at **50**

$$= \frac{50}{50 + 50}$$

50%

$$r = \frac{\boxed{s}\ \text{surplus stays at } \mathbf{50}}{\boxed{c}\ \boxed{v}} = \frac{50}{100 + 50}$$

33.3%

Capital and the rate of profit

$$r = \frac{\boxed{s}\ \text{surplus stays at } \mathbf{50}}{\boxed{c}\ \boxed{v}} = \frac{50}{200 + 50}$$

20%

The falling rate of profit

$$r = \frac{s}{c+v} = \frac{s/v}{c/v+1} = \frac{e}{1+q}$$

- $e = s/v$ is the rate of surplus value; $q = c/v$ is the composition of capital.
- Hold **v** and **s** fixed: more **c** means a larger capital advance for the same surplus.
- Hold e fixed: the higher q, the lower r.

Marx (1894), vol. III, ch. 13. Illustrative values.

Counteracting influences on the rate of profit

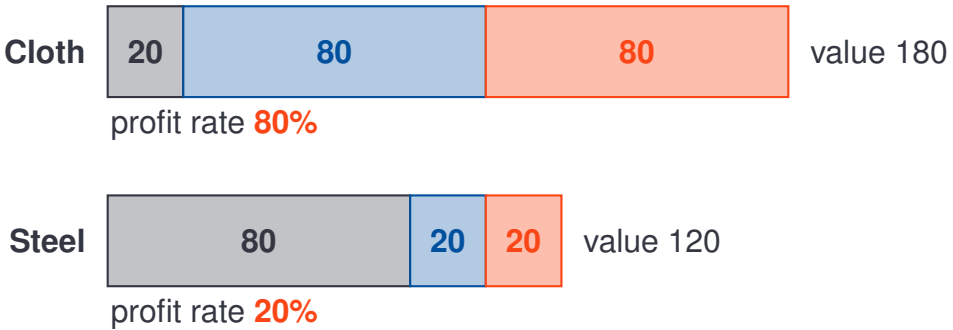
- Marx called the fall a *tendency*, checked by "counteracting influences".

Marx's six counteracting influences	
Longer, harder working days	Wages pushed below their value
Machines and materials get cheaper	Jobless workers keep labor cheap
Cheaper imported inputs and food	Railway companies settle for low dividends

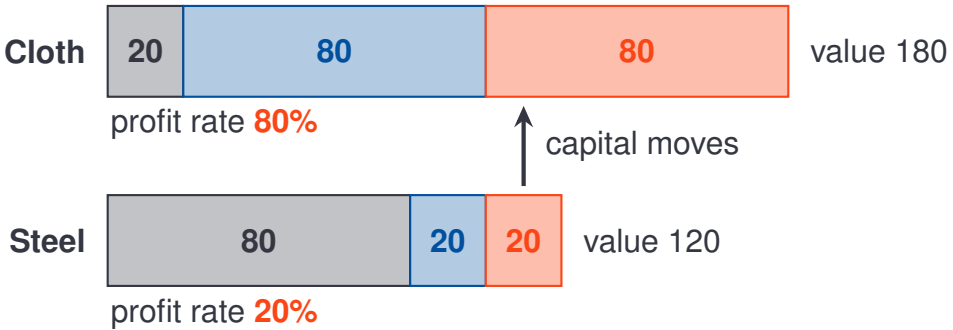
- The formula is arithmetic; the fall is a prediction. If longer, harder days raise e to 250%, the last example (**200**, **50**, **125**) is back at $r = 2.5/(1+4) = 50\%$.

Marx (1894), vol. III, chs. 3 and 13–14.

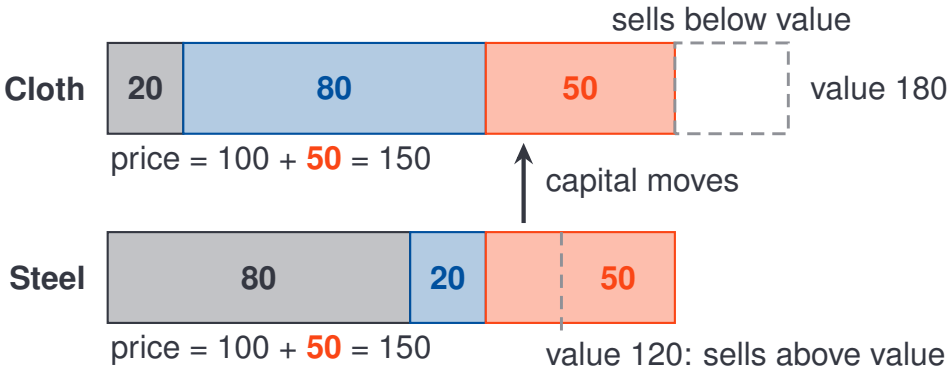
Cloth vs. steel



Equalizing profit rates



Equalizing profit rates



Capitalists as stockholders

Marx, *Capital*, vol. III, 1894

"So far as profits are concerned, the various capitalists are just so many stockholders in a stock company in which the shares of profit are uniformly divided per 100, so that profits differ [. . .] only in accordance with the amount of capital invested by each [. . .]."

New York curb brokers, c. 1906

Marx's answer: the totals

- Price of production: what it costs to make (**c** + **v**), plus profit at the average rate. Cloth and steel: 100 + **50** = 150 each.
- Cloth sells 30 below its value, steel 30 above: the deviations cancel.
- Total price 300 = total value 300. Total profit **100** = total surplus value **100**.

Marx (1894), vol. III, chs. 9–10.

▸ Marx's own table

Marx's laws of motion

Marx, *Capital*, vol. I, 1867

"[I]t is the ultimate aim of this work, to lay bare the economic law of motion of modern society"

Prediction	Where Marx says it
Workers' living standards	Manifesto (1848); *Capital* I, ch. 25
Wages vs. profits	*Wage Labour and Capital* (1849), ch. 8
The length of the working day	*Capital* I, ch. 10
The falling rate of profit	*Capital* III, chs. 13–14
Revolution in the most advanced countries	Manifesto; *Capital* I, ch. 32 and Preface

Böhm-Bawerk and the Close of Marx's System

Why does old wine cost more?

Old wine: same labor, higher price

- A cask of young wine sells for £10; after a dozen years in the cellar, £20.
- No labor went in during storage. Whose labor, then, was exploited?

Böhm-Bawerk (1884), pp. 364–365; Böhm-Bawerk (1889), p. 237.

Old wine: same labor, higher price

- A cask of young wine sells for £10; after a dozen years in the cellar, £20.
- No labor went in during storage. Whose labor, then, was exploited?
- Böhm-Bawerk: present goods are, as a rule, worth more than future goods; interest pays for the wait.

Böhm-Bawerk (1884), pp. 364–365; Böhm-Bawerk (1889), p. 237.

A note on Eugen von Böhm-Bawerk, 1851–1914

- Professor at Innsbruck (from 1881), later at Vienna (1904–14).
- Austrian finance minister three times: 1895, 1897–98, 1900–04.
- *Capital and Interest* (2 vols., 1884, 1889): why capital earns interest.
- *Karl Marx and the Close of His System* (1896).
- On Austria's 100-schilling note (dated 1984; in use 1985–2002).

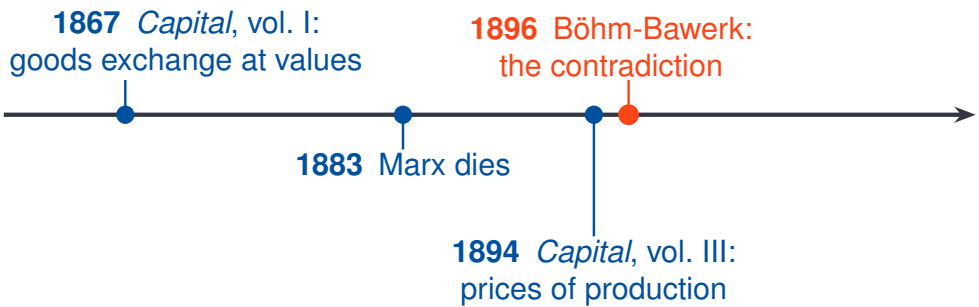
Böhm-Bawerk (1851–1914)

Zwiedineck-Südenhorst (1955).

From Volume I to Volume III

Law of value: goods exchange in proportion to the labor time needed to make them.

Transformation problem: getting from values (Volume I) to prices of production (Volume III).



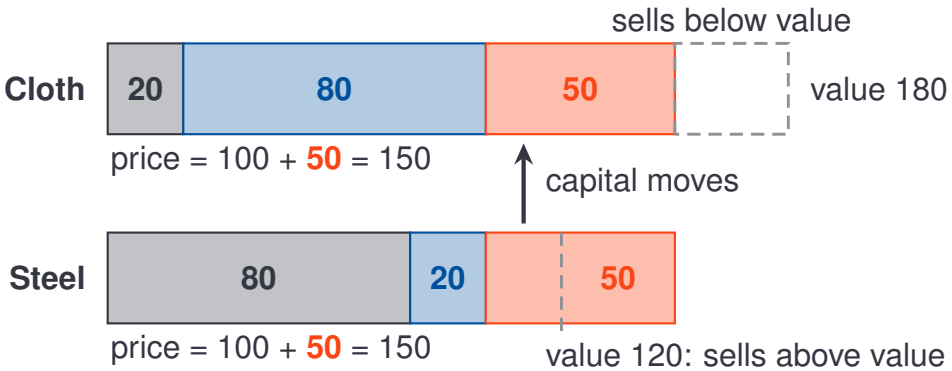
Böhm-Bawerk's verdict on Volume III

Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"I cannot help myself; I see here no explanation and reconciliation of a contradiction, but the bare contradiction itself. Marx's third volume contradicts the first."

Böhm-Bawerk
(1851–1914)

The contradiction in numbers



- Volume I, at values: cloth : steel = 180 : 120.
- Volume III, at prices of production: 150 : 150. They cannot both hold.

Elephants and mayflies

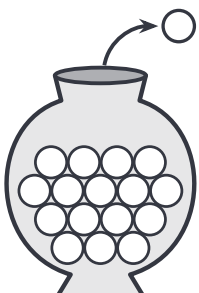
Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"[E]lephants live on an average one hundred years and May flies only a single day, [. . .] we can strike an average of fifty years. [. . .] The deviations from this average 'mutually cancel each other,' and consequently on the whole and on the average the law that all kinds of animals have the same length of life is established!"

Böhm-Bawerk's urn

Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"He acts as one who urgently desiring to bring a white ball out of an urn takes care to secure this result by putting in white balls only."



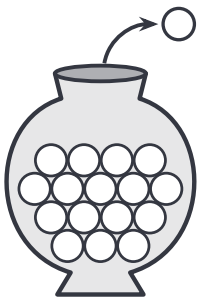
only white balls went in

white = made by labor

Böhm-Bawerk's urn

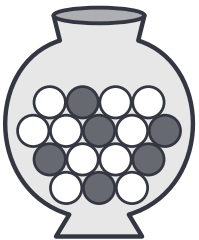
Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"He acts as one who urgently desiring to bring a white ball out of an urn takes care to secure this result by putting in white balls only."



only white balls went in

white = made by labor



everything that is exchanged

dark = gifts of nature: soil, coal, water power

Marx on usefulness and exchange value

Marx, *Capital*, vol. I, 1867

"As use-values, commodities are, above all, of different qualities, but as exchange values they are merely different quantities, and consequently do not contain an atom of use-value."

- A coat keeps us warm; bread feeds us. These are different uses.
- Marx sets aside those different uses to seek a common measure of how much one good exchanges for another.

Marx's common factor: abstract labor

Marx, *Capital*, vol. I, 1867

"If then we leave out of consideration the use-value of commodities, they have only one common property left, that of being products of labour."

- Marx also sets aside the differences between tailoring and baking: both count as abstract human labor.
- Böhm-Bawerk's objection: why keep labor in the abstract after rejecting usefulness in the abstract?

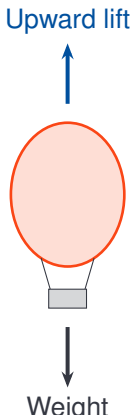
Marx on supply and demand

Marx, *Capital*, vol. III, 1894, ch. 10

"If supply and demand balance one another, they cease to explain anything, do not affect market-values"

- Marx: shortages and surpluses explain why prices move away from their normal level.
- But when supply equals demand, what explains that level? Why does the balance occur at one price rather than another?
- He looks to the conditions of production to explain the normal price.

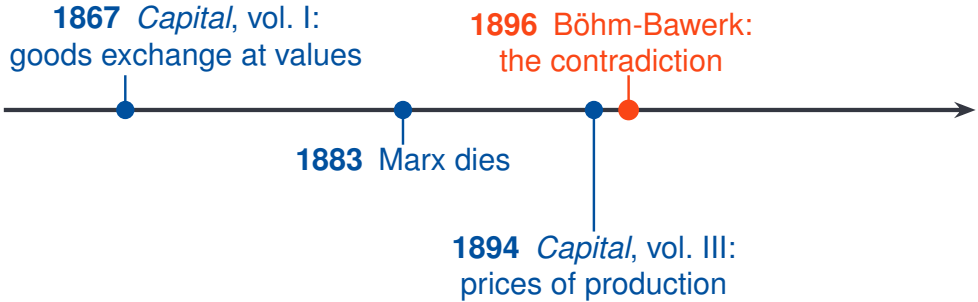
Böhm-Bawerk's balloon



- A balloon can float at rest because upward lift balances its weight. Both forces still act.
- Their balance explains where it floats. Change the conditions and the balance changes.
- Likewise, buyers' willingness to buy and sellers' willingness to sell explain where prices settle.

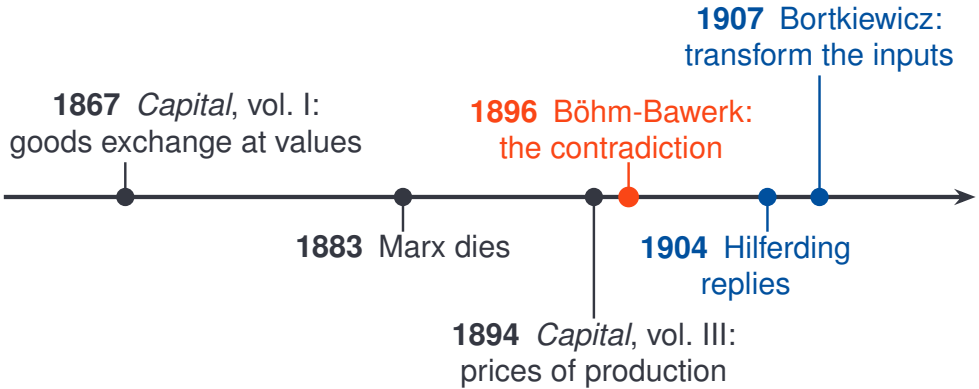
Böhm-Bawerk (1896), ch. 4.

The transformation problem after Böhm-Bawerk



Duménil and Foley (2008).

The transformation problem after Böhm-Bawerk



Duménil and Foley (2008).

Marx's two aggregate equalities

Individual goods can sell above or below their labor values. Marx argues that these differences cancel across the economy.

$$\text{Total selling prices} = \text{Total labor values}$$

$$\text{Total profits} = \text{Total surplus value}$$

- Competition redistributes surplus value among capitalists; it does not create an extra pool of value.
- These are claims about economy-wide totals, not equality for each commodity.

Marx (1894), vol. III, chs. 9–10. Labor values expressed in the same monetary units as prices.

Bortkiewicz's objection

- One firm's selling price is another firm's input cost: both must be adjusted together.
- Bortkiewicz showed that, with consistent input and output prices, Marx's two aggregate equalities need not both hold.
- Equal profit rates therefore do not, by themselves, preserve Marx's totals.

Ladislaus von Bortkiewicz (1868–1931)

Bortkiewicz (1907), pp. 201–206.

What survives of the labor theory of value?

- Labor is one cause of prices, not the only one:
  wages, capital and waiting move them too.
- Marx's question survives: where does profit come from?
- Böhm-Bawerk's answer is time, as in the wine:
  present goods are, as a rule, worth more than future goods.

Pareto’s Critique

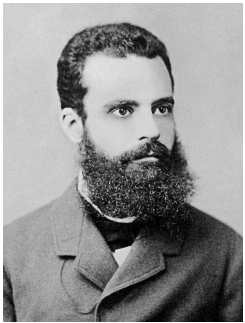
Capital vs. the capitalist

Pareto, Introduction to Marx's *Capital*, 1893

"Marx's book should be entitled The Capitalist *rather than* Capital*, at least if one takes the latter word in the fairly generally accepted sense of economic goods intended to facilitate the production of other goods."*

A note on Vilfredo Pareto, 1848–1923

- Born in Paris, 1848; civil-engineering degree, Turin, 1870.
- Railway engineer, then ironworks manager: general manager of the Società delle Ferriere Italiane, 1880–1890.
- Succeeded Walras in the chair of political economy at Lausanne, 1893.
- "Pareto's law" of incomes (*Cours*, 1896–97); "Pareto efficiency" (*Manuale*, 1906).
- Critic of Marx: Introduction to *Capital* (1893), *Les Systèmes socialistes* (1902–03), *Trattato di sociologia generale* (1916).



Vilfredo Pareto
(1848–1923)

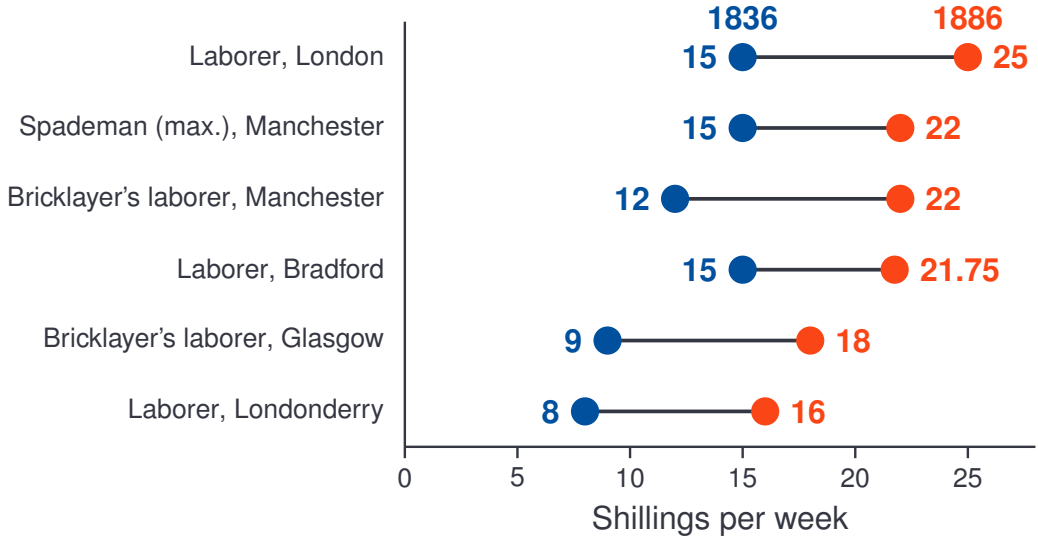
Mornati (2014).

Who creates the surplus value?

Pareto's sewing-machine example

- A seamstress pays 30 centimes to rent a machine and needs 70 centimes to live: 100 centimes to cover.
- Three hours of sewing cover that amount. She works six hours, so the remaining receipts are hers.
- Both her work and the machine help produce those receipts. Having money left over does not, by itself, prove that labor alone created it.

Pareto (1893), pp. XXXIII–XXXIV. Explanation of his example.

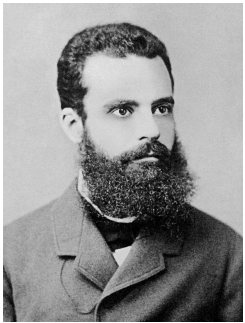


Note: money (not real) wages of unskilled non-farm labor, "fifty years ago" and "the present time" (about 1883–84), which Pareto labels 1836 and 1886. Consumer prices were roughly flat. Giffen (1886), p. 425, as printed in Pareto (1893), p. LXVI.

Elites vs. classes

Elite: the few who hold power, such as nobles, generals or party leaders.

- Pareto (1902), pp. 117–118, credits Marx with drawing attention to class struggle as a real force in history.
- Pareto: conflict runs among many groups, especially rival elites competing for power.
- The winners of a revolution become a new elite
  $\Rightarrow$ Prediction for a socialist revolution: a new ruling class.

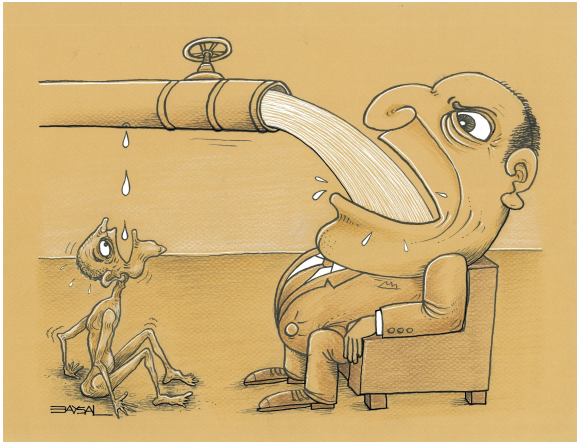


Vilfredo Pareto
(1848–1923)

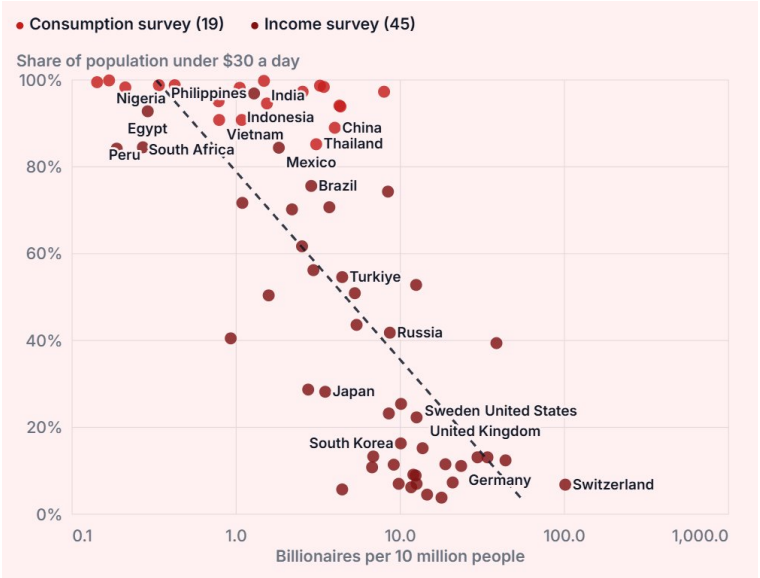
Pareto (1902), pp. 35–36, 117–118.

Marx’s Predictions

Does billionaire wealth come at the expense of the poor?



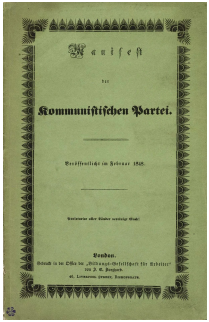
Ercan Baysal, *Injustice* (2010).



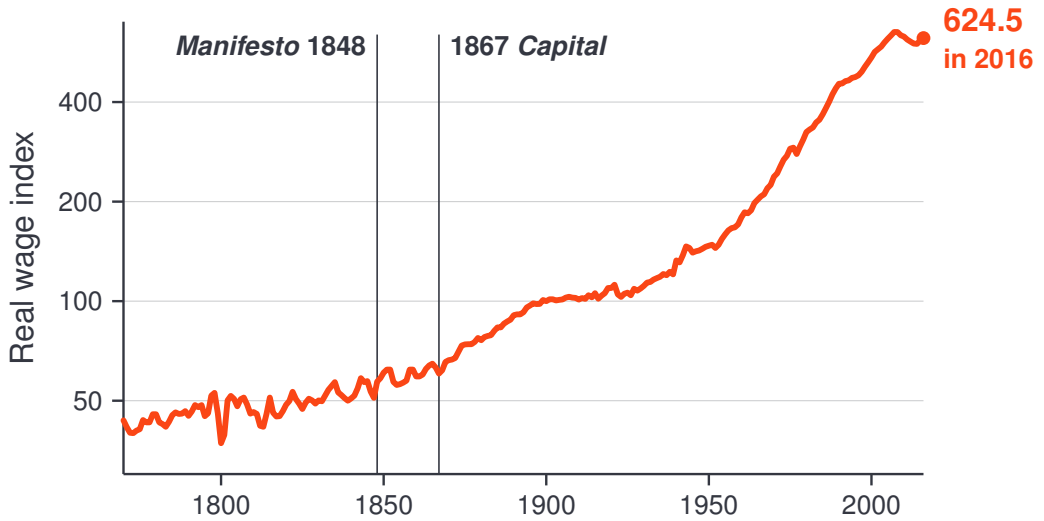
Note: 64 countries; $30/day in 2021 PPP dollars; latest available surveys, mixed years. Forbes 2026 and World Bank PIP, reproduced from Clarke (2026). Correlation, not a causal estimate.

Marx and Engels, *Manifesto of the Communist Party*, 1848

"The modern labourer, on the contrary, instead of rising with the progress of industry, sinks deeper and deeper below the conditions of existence of his own class."


Manifest
der
Kommunistischen Partei.
London.

Manifest der Kommunistischen Partei
London, February 1848



Note: average real earnings in Britain, index 1900 = 100, log scale; Feinstein (1998) before 1880. Thomas and Dimsdale (2017).

Absolute vs. relative poverty

Marx, *Wage Labour and Capital*, 1849

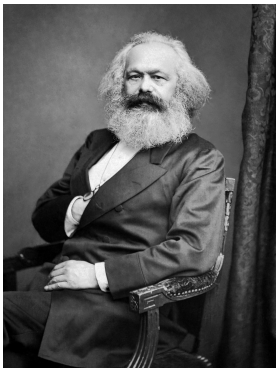
"A house may be large or small; as long as the neighboring houses are likewise small, it satisfies all social requirement for a residence. But let there arise next to the little house a palace, and the little house shrinks to a hut."

Doré, *London* (1872)

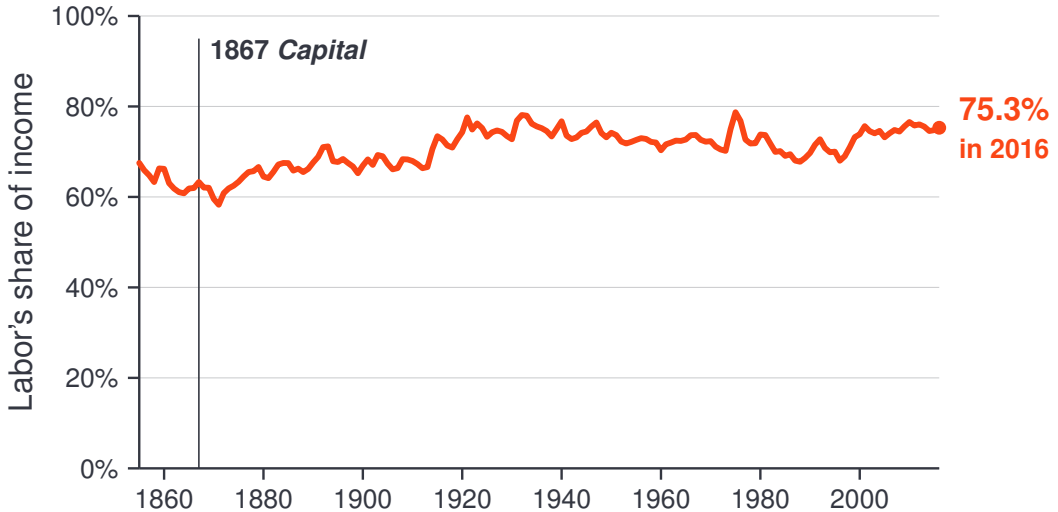
Wages vs. profits

Marx, *Wage Labour and Capital*, 1849

"If capital grows rapidly, wages may rise, but the profit of capital rises disproportionately faster."



Karl Marx (1818–1883)



Note: labor's share of UK national income, excluding housing rents. Thomas and Dimsdale (2017).

Wages vs. profits: the record

- Allen (2009) estimates: the profit rate doubled from 1800 to 1840, and capital's share rose from 20% (late 1700s) to over 40% (mid-1800s).
- Kaldor (1961), summing up the record to his day: in developed capitalist economies, a steady rate of profit and steady shares of profits and wages.

Allen (2009); Kaldor (1961).

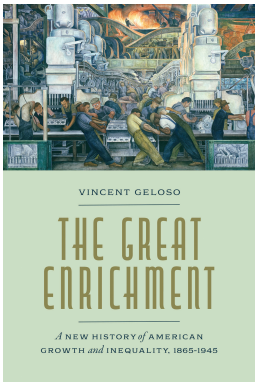
Measuring Britain's labor share

- Crafts (2020) uses revised output data and prices of what Britain produced, rather than what workers bought.
- His estimate of labor's share in 1860: 60%, compared with 44% in Allen. Both start near 56% in 1800.
- Capital's share is what remains after wages and land rents. Changes in those estimates change the inferred profit rate.

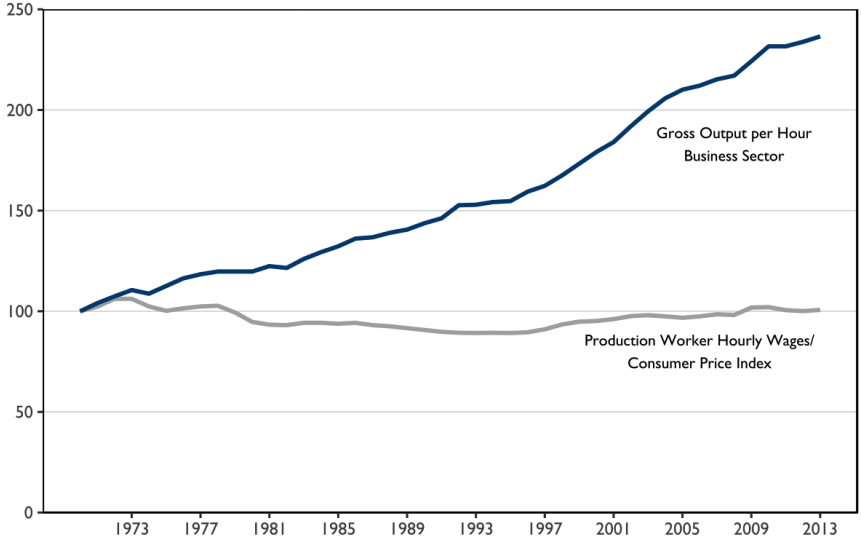
Crafts (2020), Tables 2–4.

Inequality and the cost of living

- Geloso studies the US, 1865–1945: different households buy different baskets.
- Poor households spent a larger share of their budgets on food. Cheaper staples could improve their purchasing power faster than incomes alone suggest.
- A rising gap in dollar incomes need not imply the same rise in the gap in living standards.



Geloso, *The Great Enrichment* (forthcoming 2027), author preview, ch. 2.



Note: US, 1970–2013; 1970 = 100. Gross business output/hour vs. production-worker cash wages/hour in consumer prices. Lawrence (2016), reproduced in Strain (2019), p. 175.

Productivity and pay: workers and compensation

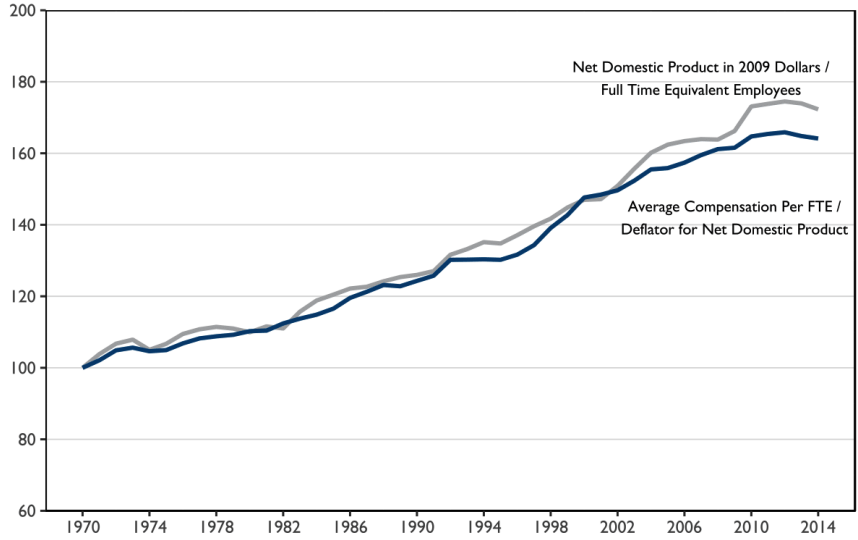
- Match the workers: compare economy-wide production with average compensation for all workers, including managers.
- Count benefits: employer health insurance and pension contributions pay workers too. More benefits can mean slower growth in cash wages.
- Pay for a particular group tells us how gains are distributed among workers. Its gap with productivity is not automatically income captured by capital.

Strain (2019), pp. 171–175; de Rugy (2022).

Productivity and pay: prices and depreciation

- Use output prices: a baker is paid out of bread sales. Compare the baker's pay with the price of bread.
- With bread prices fixed, output per hour and hourly pay can rise 10% while rent rises 20%. Pay tracks productivity, yet buys less housing: purchasing power is a different question.
- Subtract depreciation: replacing worn-out equipment uses part of output. That part is not income to divide between labor and capital. Use *net* output.

Strain (2019), pp. 172–175; de Rugy (2022).



Note: US, 1970–2014; 1970 = 100. Net output and average total compensation per full-time-equivalent worker (FTE), both in output prices. Lawrence (2016), reproduced in Strain (2019), p. 176.

The link between productivity and pay

- With comparable measures, average compensation tracks productivity much more closely in Lawrence's comparison.
- Stansbury and Summers: 1 percentage point faster productivity growth is associated with about 0.7–1 point faster median compensation growth, 1973–2016.
- Productivity can push pay up while other forces hold it back. A gap between the two lines does not show that productivity stopped raising pay.

Strain (2019); Stansbury and Summers (2017).

The length of the working day

- Marx: capital is “reckless of the health or length of life of the labourer, unless under compulsion from society”.
- The Factory Acts:
  - 1833: in textile mills (silk excepted), no children under 9; ages 9–13 at most 9 hours a day.
  - 1847: 10 hours a day for women and teenagers.
  - 1850: work only between 6 a.m. and 6 p.m., 10½ hours Monday to Friday, 7½ on Saturday: 60 hours a week.

Factory Acts (1833, 1847); Marx (1867), vol. I, ch. 10.

Testing a conditional prediction

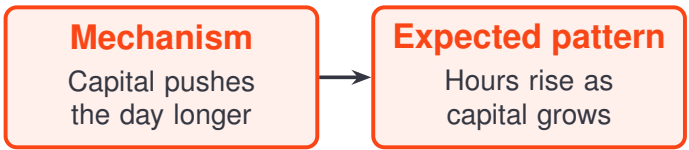
Conditional prediction: a force produces a pattern unless stronger forces push back.

Mechanism

Capital pushes the day longer

Testing a conditional prediction

Conditional prediction: a force produces a pattern unless stronger forces push back.

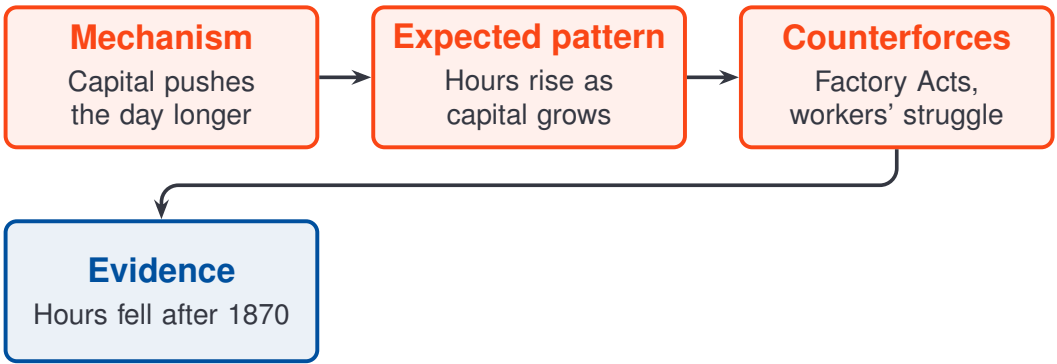


Testing a conditional prediction

Conditional prediction: a force produces a pattern unless stronger forces push back.

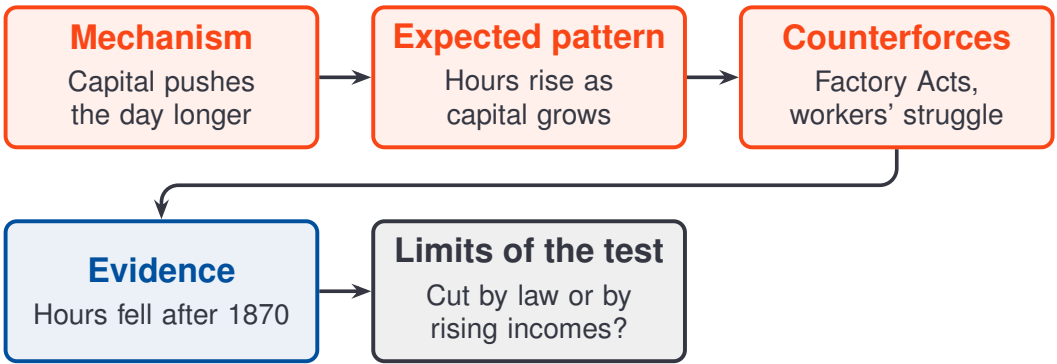
Testing a conditional prediction

Conditional prediction: a force produces a pattern unless stronger forces push back.



Testing a conditional prediction

Conditional prediction: a force produces a pattern unless stronger forces push back.



Marx: competition and the working day

- Suppose one employer cuts the day from 12 to 10 hours at the same daily pay. It now pays more per hour, while rivals keep the longer day.
- In West's reading of Marx, competition punishes the employer acting alone, even if shorter hours would protect the workforce over time.
- A common legal limit makes all employers shorten the day together. This is Marx's case for compulsion.

West (1983), pp. 267–268, interpreting Marx, *Capital* I, ch. 10.

West: competition for workers

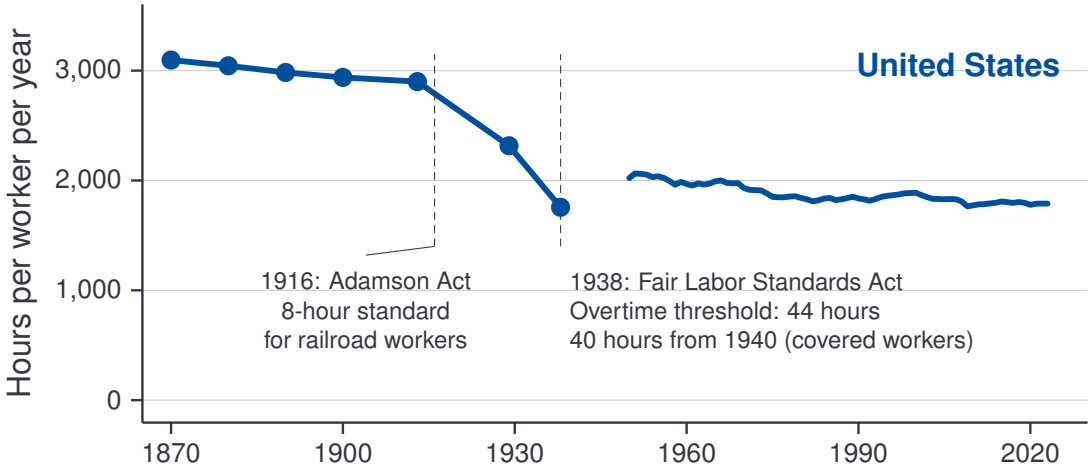
- Workers care about both pay and free time. As incomes rise, they may prefer shorter hours to still more earnings.
- Employers compete for workers too. A firm offering shorter hours can attract or retain workers who value that arrangement.
- Competition can therefore support shorter hours before a common legal limit. Hours and pay are negotiated together.

West (1983), pp. 272, 275–277.

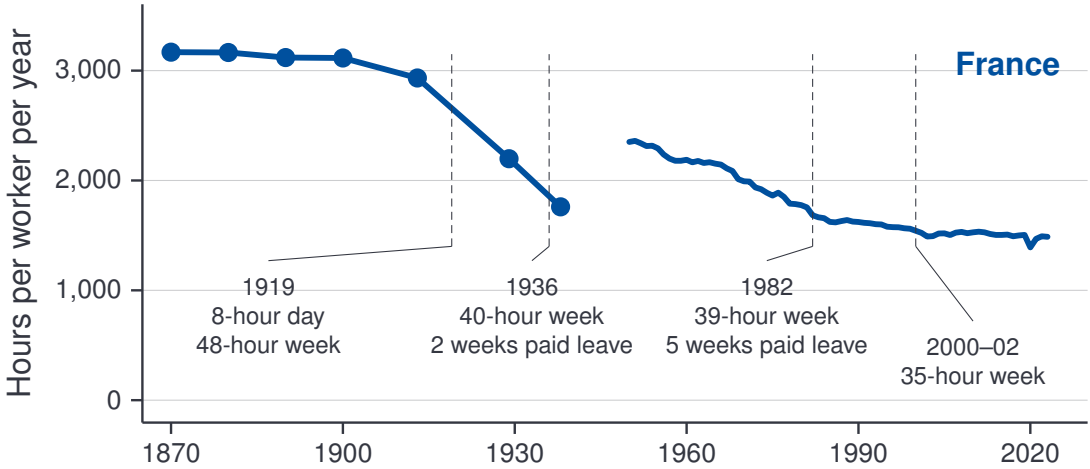
West: the workers' choices

- Factory inspector Horner asked workers in 1848 to choose: 10 hours with less pay, or longer hours with more pay.
- Among 1,153 workers, about 62% chose 10 hours; 25% chose 12. Half of the lowest-paid men wanted more than 10 hours.
- West's point: workers did not all want the same bargain. A shorter day had an income cost, which mattered more to some workers.

West (1983), pp. 274–277, reexamining Horner's report.



Note: annual hours, Huberman–Minns / PWT 11, via OWID; worker coverage changes across the gap. Legal weekly standards: historical sources. Dates do not identify effects.



Note: annual hours, Huberman–Minns / PWT 11, via OWID; worker coverage changes across the gap. Legal weekly standards: historical sources. Dates do not identify effects.

What would a falling rate of profit look like?

$$r = \frac{s}{c+v} = \frac{e}{1+q}$$

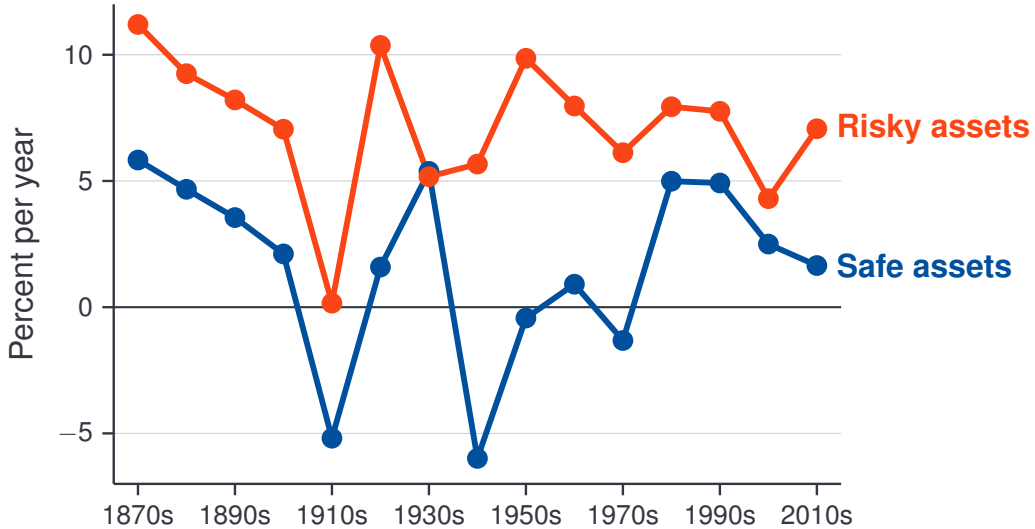
r: the rate of profit, profit per dollar advanced. $e = s/v$: the rate of surplus value.
$q = c/v$: the composition of capital, machines and materials per dollar of wages.

What would a falling rate of profit look like?

$$r = \frac{s}{c+v} = \frac{e}{1+q}$$

r: the rate of profit, profit per dollar advanced. $e = s/v$: the rate of surplus value.
$q = c/v$: the composition of capital, machines and materials per dollar of wages.

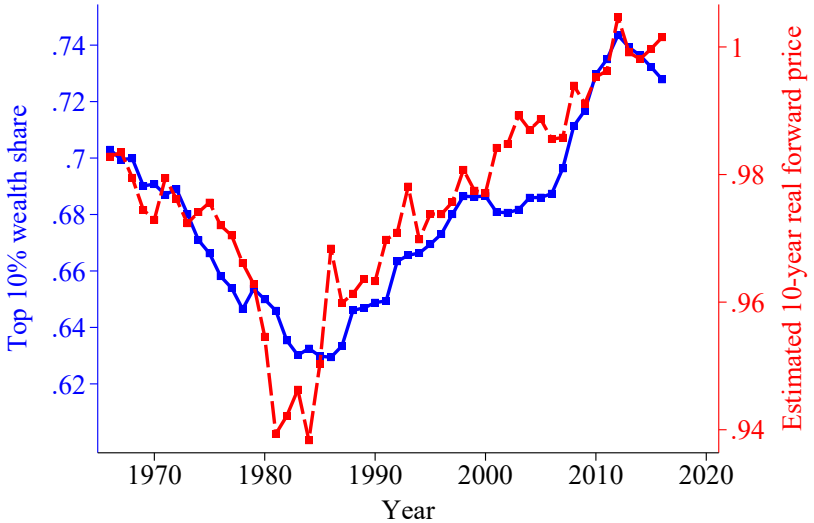
- Mechanism: competition pushes every mill to mechanize, so q climbs.
- Expected pattern: with e fixed at 100%, q rising from 1 to 4 cuts r from 50% to 20% (the expanding-capital example). A long downward drift.
- Counterforces: e rises, or machines get cheaper and q stops climbing.
- Evidence: the rate of return, what a dollar in shares and houses (risky) or government bonds (safe) earns in a year, after inflation.



Note: real returns (after inflation), decade averages across 16 rich countries weighted by GDP; before 1890 the risky series covers only two to five countries. Risky: equities and housing. Safe: government bonds and bills. Jordà et al. (2019).

Measuring Marx's rate of profit

- Market returns on shares and houses show little sign of a long downward drift (Jordà et al. 2019).



Note: US top 10% wealth share (blue). Red is one minus the estimated 10-year real forward rate: **up means lower interest rates**. Catherine et al. (2026), Fig. 3B, using Piketty, Saez and Zucman (2018).

Why did the revolution come in Russia?

Russia in 1917

Engels (1847): the revolution would come simultaneously in all civilized countries, at least England, America, France and Germany.

- 1897 census: about three in four workers were in agriculture (74.6%).
- Serfdom abolished in 1861; peasants got land in communal, not individual, ownership.
- Industry grew about 7% a year in the 1890s, faster than Germany or the U.S.
- The Putilov works in Petrograd employed more than 24,000 workers.

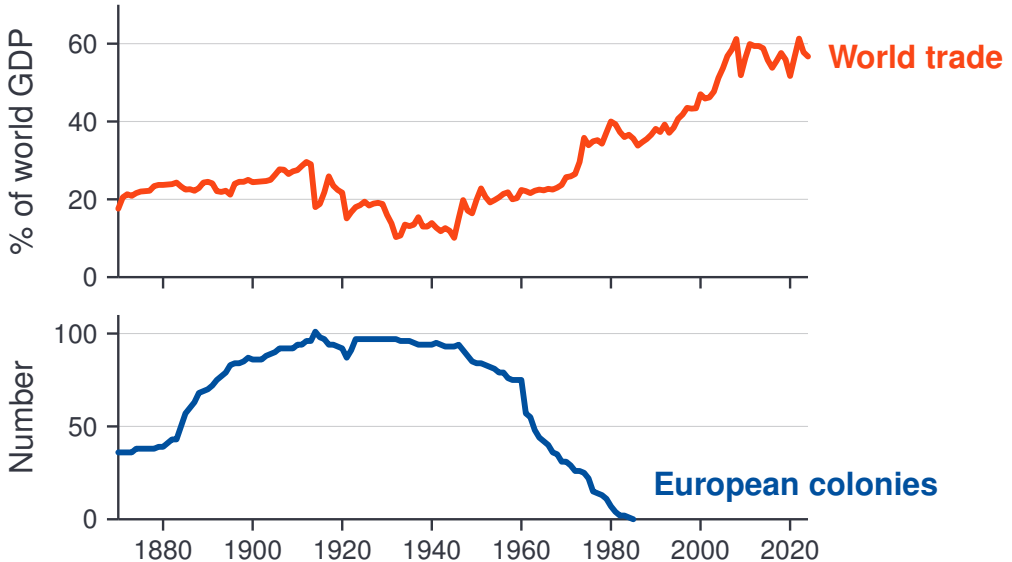
$\Rightarrow$ Not wholly feudal, but mostly peasant.

Engels (1847), Q. 19; Thorp and Thorp (1926), p. 235; Fischer (1994); Hasegawa (1981).

Lenin's revisions of Marx

- *What Is to Be Done?* (1902): workers alone reach only "trade union consciousness"; a party of professional revolutionaries must lead.
- *Imperialism* (1916): "imperialism is the monopoly stage of capitalism"; capitalism survives by exporting capital and dividing the world.

V. I. Lenin
(1870–1924)

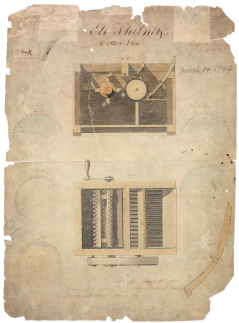


Note: world exports plus imports; to 1949, Klasing and Milionis (2014), higher estimate; then Penn World Table and World Bank.
Colonies: today's countries then ruled by eight European powers (Becker 2023). Via Our World in Data.

Slavery and the cotton market

Marx, *Capital*, vol. I, 1867

"[A]s soon as people, whose production still moves within the lower forms of slave-labour [. . .] are drawn into the whirlpool of an international market [. . .] the civilized horrors of over-work are grafted on the barbaric horrors of slavery, serfdom, &c."



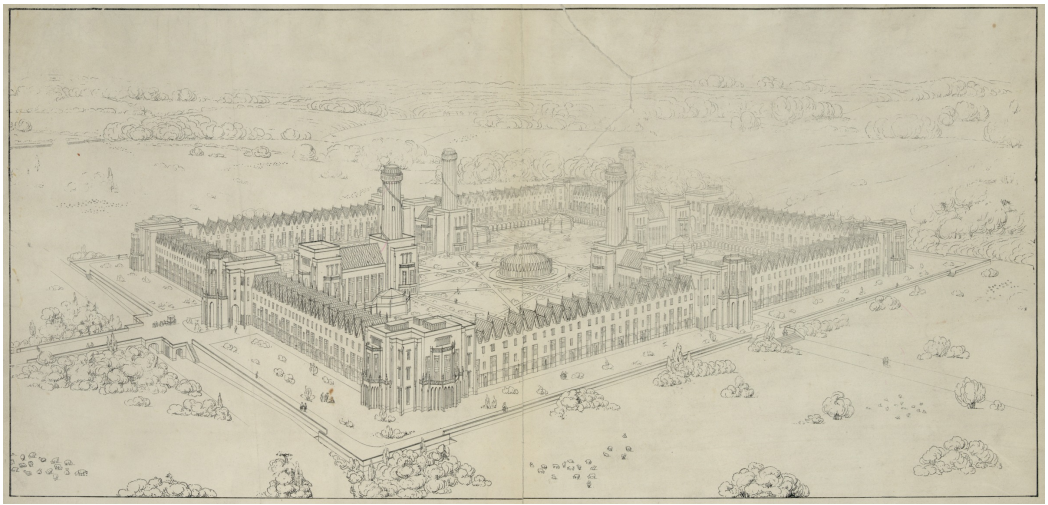
Whitney's cotton gin patent, 1794

Slavery: what the record shows

- Abolition: Britain 1833 (in force 1834), France 1848, U.S. 1865, Cuba 1886, Brazil 1888.
- Enslaved people were about 15–20% of U.S. wealth in 1860, and 40–50% of Southern wealth.

Dray et al. (2026), pp. 27–28; Wright (2006), p. 60.

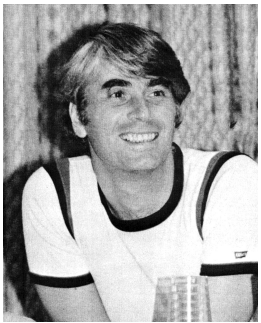
Was Owen free to try his utopia in America?



Nozick on socialism and freedom

Nozick, *Anarchy, State, and Utopia*, 1974

"The socialist society would have to forbid capitalist acts between consenting adults."



Robert Nozick
(1938–2002)

Marx's own example: five spheres

Marx, *Capital* III, ch. 9: every capital = 100, e = 100%, all **c** used up.

Sphere	c + v	s	Value	Price	Deviation
I	80c + 20v	20	120	122	+2
II	70c + 30v	30	130	122	−8
III	60c + 40v	40	140	122	−18
IV	85c + 15v	15	115	122	+7
V	95c + 5v	5	105	122	+17
Total	**500**	**110**	**610**	**610**	**0**

- Average rate of profit: **110** / 500 = 22%, so each capital of 100 earns 22.
- Marx: each fifth of the product sells at 122. Deviations cancel: +26 and −26.

Back

The wage rise, step by step

	c	v	s	Value	Capital	Price
Cloth	20	80 → 88	80 → 72	180	108	154.29
Steel	80	20 → 22	20 → 18	120	102	145.71

1. Same workers, same hours: the new value each industry creates (**v** + **s**) stays 160 in cloth and 40 in steel. Wages rise 10%, so **v** takes more of it and **s** shrinks.
2. Average rate of profit: **90** of surplus value on a capital (**c** + **v**) of 108 + 102 = 210, so $r = 3/7 \approx 42.9\%$.
3. Price = capital $\times (1 + 3/7)$ = capital $\times 10/7$: cloth 154.29, steel 145.71, total 300. Values stay 180 and 120; before the rise, both prices were 150.

Back to the discussion

References (1/24)

Allen, Robert C. (2001). "The Great Divergence in European Wages and Prices from the Middle Ages to the First World War." *Explorations in Economic History* 38(4), 411–447.

Allen, Robert C. (2009). "Engels' Pause: Technical Change, Capital Accumulation, and Inequality in the British Industrial Revolution." *Explorations in Economic History* 46(4), 418–435.

Angell, Norman (1910). *The Great Illusion*. New York: G. P. Putnam's Sons. Preface to the fourth American edition (1913), p. vii.

Basu, Deepankar and Panayiotis T. Manolakos (2013). "Is There a Tendency for the Rate of Profit to Fall? Econometric Evidence for the U.S. Economy, 1948–2007." *Review of Radical Political Economics* 45(1), 76–95.

References (2/24)

Basu, Deepankar, Julio Huato, Jesus Lara Jauregui and Evan Wasner (2025). "World Profit Rates, 1960–2019," *Review of Political Economy* 37(1), 92–107.

Becker, Bastian (2023). *Colonial Dates Dataset (COLDAT)*, version 3.0. Country-level dates of European colonial rule; series accessed through Our World in Data.

***Birmingham Daily Gazette* (1901).** Description of industrial housing, quoted in Jesús Fernández-Villaverde, *Karl Marx*, printed slide 14 (PDF p. 17). Original newspaper issue not inspected.

Blaug, Mark (1997). *Economic Theory in Retrospect*, 5th ed. Cambridge University Press, p. 215. Quoted in Jesús Fernández-Villaverde, *Karl Marx*, printed slide 8 (PDF p. 10); original book page not inspected.

References (3/24)

Böhm-Bawerk, Eugen von (1884). *Capital and Interest: A Critical History of Economical Theory*. Trans. William Smart. London: Macmillan, 1890.

Böhm-Bawerk, Eugen von (1889). *The Positive Theory of Capital*. Trans. William Smart. London: Macmillan, 1891.

Böhm-Bawerk, Eugen von (1896). *Karl Marx and the Close of His System*. Trans. Alice M. Macdonald (1898), ed. Paul M. Sweezy. New York: Augustus M. Kelley, 1949.

Bortkiewicz, Ladislaus von (1907). "On the Correction of Marx's Fundamental Theoretical Construction in the Third Volume of *Capital*." Trans. Paul M. Sweezy, in Sweezy (ed.), *Karl Marx and the Close of His System*. New York: Augustus M. Kelley, 1949, 199–221.

References (4/24)

Cambridge Group for the History of Population and Social Structure (n.d.). *Mortality in Manchester*. University of Cambridge.

Children's Employment Commission (1842). *First Report of the Commissioners: Mines*. London: W. Clowes and Sons, p. 71, para. 306. Sarah Gooder's testimony, quoting Appendix, Part I, evidence no. 116.

Clark, Gregory (2005). "The Condition of the Working Class in England, 1209–2004." *Journal of Political Economy* 113(6), 1307–1340.

Crafts, Nicholas and Terence C. Mills (2022). "Considering the Counterfactual: Real Wages in the First Industrial Revolution," *Economic Journal* 132(645), 1994–2006.

References (5/24)

Davenport, Romola J. (2020). “Urbanization and Mortality in Britain, c. 1800–50,” *Economic History Review* 73(2), 455–485.

Davenport, Romola J. (2021). “Nineteenth-Century Mortality Trends: A Reply to Szreter and Mooney.” *Economic History Review* 74(4), 1096–1110.

Davis, Lance E. and Robert A. Huttenback (1986). *Mammon and the Pursuit of Empire: The Political Economy of British Imperialism, 1860–1912*. Cambridge University Press.

Djilas, Milovan (1957). *The New Class: An Analysis of the Communist System*. New York: Praeger, p. 36.

References (6/24)

Dray, Sacha, Camille Landais and Stefanie Stantcheva (2026). "Wealth and Property Taxation in the United States," NBER Working Paper 31080, pp. 27–28.

Duménil, Gérard and Duncan K. Foley (2008). "The Marxian Transformation Problem," in *The New Palgrave Dictionary of Economics*, 2nd ed. Basingstoke: Palgrave Macmillan.

Engels, Friedrich (1847). *The Principles of Communism*, Q. 19.

Engels, Friedrich (1880). *Socialism: Utopian and Scientific*. English edition, trans. Edward Aveling. London: Swan Sonnenschein, 1892.

References (7/24)

Engels, Friedrich (1885). Preface to Karl Marx, *Capital*, vol. II, 5 May 1885.

Feinstein, Charles H. (1998). “Pessimism Perpetuated: Real Wages and the Standard of Living in Britain during and after the Industrial Revolution.” *Journal of Economic History* 58(3), 625–658.

Fischer, Stanley (1994). “Russia and the Soviet Union Then and Now.” In O. J. Blanchard, K. A. Froot and J. D. Sachs (eds.), *The Transition in Eastern Europe*, vol. 1. University of Chicago Press, 221–258.

Giffen, Robert (1886). “Further Notes on the Progress of the Working Classes in the Last Half Century.” *Journal of the Statistical Society of London* 49(1), 28–91. Reprinted in *Essays in Finance*, Second Series, pp. 424–425.

References (8/24)

Great Britain (1833, 1847). *Factory Act 1833*, 3 & 4 Will. IV c. 103, ss. 7–8 and 17; *Factories Act 1847* (Ten Hours Act), 10 & 11 Vict. c. 29, ss. I–II.

Griffin, Emma (2013). *Liberty's Dawn: A People's History of the Industrial Revolution*. Yale University Press.

Hasegawa, Tsuyoshi (1981). *The February Revolution: Petrograd, 1917*. University of Washington Press.

Heldring, Leander, James A. Robinson and Sebastian Vollmer (2024). *The Economic Effects of the English Parliamentary Enclosures*. Working paper, May version, section 9 and Table 7; Appendix Tables A15–A17.

References (9/24)

Hilferding, Rudolf (1904). “Böhm-Bawerk’s Criticism of Marx.” Trans. Eden and Cedar Paul, in Paul M. Sweezy (ed.), *Karl Marx and the Close of His System*. New York: Augustus M. Kelley, 1949, 121–196.

Huberman, Michael and Chris Minns (2007). “The Times They Are Not Changin’: Days and Hours of Work in Old and New Worlds, 1870–2000.” *Explorations in Economic History* 44(4), 538–567.

Humphries, Jane (2013). “Childhood and Child Labour in the British Industrial Revolution,” *Economic History Review* 66(2), 395–418.

Jordà, Òscar, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick and Alan M. Taylor (2019). “The Rate of Return on Everything, 1870–2015.” *Quarterly Journal of Economics* 134(3), 1225–1298. Data: JST Macrohistory Database, R6.

References (10/24)

Kaldor, Nicholas (1961). “Capital Accumulation and Economic Growth,” in Lutz and Hague (eds.), *The Theory of Capital*, 177–222.

Karabarbounis, Loukas and Brent Neiman (2014). “The Global Decline of the Labor Share,” *Quarterly Journal of Economics* 129(1), 61–103.

Klasing, Mariko J. and Petros Milionis (2014). “Quantifying the Evolution of World Trade, 1870–1949.” *Journal of International Economics* 92(1), 185–197.

Lenin, V. I. (1902). *What Is to Be Done?*, chs. II and IV. Trans. Joe Fineberg and George Hanna, in *Collected Works*, vol. 5. Moscow: Foreign Languages Publishing House, 1961.

References (11/24)

Lenin, V. I. (1916; published 1917). *Imperialism, the Highest Stage of Capitalism*, ch. VII.

Magness, Phillip W. and Michael Makovi (2023). "The Mainstreaming of Marx: Measuring the Effect of the Russian Revolution on Karl Marx's Influence." *Journal of Political Economy* 131(6), 1507–1545. Replication data: Harvard Dataverse, doi:10.7910/DVN/S14QCI.

Maito, Esteban Ezequiel (2014). "The Historical Transience of Capital: The Downward Trend in the Rate of Profit since XIX Century." MPRA Paper 55894.

Martin, Philippe, Thierry Mayer and Mathias Thoenig (2008). "Make Trade Not War?," *Review of Economic Studies* 75(3), 865–900.

References (12/24)

Marx, Karl (1844). *Economic and Philosophical Manuscripts of 1844*, “Estranged Labour.”

Marx, Karl (1847). *The Poverty of Philosophy*, ch. II, section 1, “Second Observation.”

Marx, Karl (1849). *Wage Labour and Capital*. Revised edition edited by Friedrich Engels, 1891.

Marx, Karl (1853). “The Future Results of British Rule in India.” *New-York Daily Tribune*, 8 August 1853; written 22 July 1853.

References (13/24)

Marx, Karl (1859). *A Contribution to the Critique of Political Economy*, Preface.

Marx, Karl (1861). “The North American Civil War.” *Die Presse*, no. 293, 25 October 1861. English translation in *Marx–Engels Collected Works*, vol. 19.

Marx, Karl (1867). *Capital: A Critical Analysis of Capitalist Production*, vol. I. Trans. Samuel Moore and Edward Aveling. London: Swan Sonnenschein, 1887.

Marx, Karl (1885). *Capital: A Critique of Political Economy*, vol. II. Edited by Friedrich Engels.

References (14/24)

Marx, Karl (1894). *Capital: A Critique of Political Economy*, vol. III. Edited by Friedrich Engels.

Marx, Karl and Friedrich Engels (1848). *Manifesto of the Communist Party*. English edition translated by Samuel Moore, edited by Friedrich Engels, 1888.

Marx, Karl and Friedrich Engels (1882). Preface to the Russian edition of the *Manifesto of the Communist Party*, 21 January 1882.

Mornati, Fiorenzo (2014). “Pareto, Vilfredo,” *Dizionario Biografico degli Italiani*, vol. 81. Roma: Istituto della Enciclopedia Italiana.

Nardinelli, Clark (1990). *Child Labor and the Industrial Revolution*. Indiana University Press.

Nozick, Robert (1974). *Anarchy, State, and Utopia*. New York: Basic Books, ch. 7, p. 163.

Olmstead, Alan L. and Paul W. Rhode (2008). "Biological Innovation and Productivity Growth in the Antebellum Cotton Economy," *Journal of Economic History* 68(4), 1123–1171.

Our World in Data. Grapher series for annual working hours, world trade and European colonies. Underlying sources: Huberman and Minns; Penn World Table; Klasing and Milionis; World Bank; Becker. Data used in the charts compiled October 2026.

References (16/24)

Pareto, Vilfredo (1893). “Introduction,” in Karl Marx, *Le Capital*, extraits faits par M. Paul Lafargue. Paris: Guillaumin. Sewing-machine argument, p. XXXIII; wage table, p. LXVI.

Pareto, Vilfredo (1896–97). *Cours d’économie politique*, 2 vols. Lausanne: F. Rouge.

Pareto, Vilfredo (1902–03). *Les Systèmes socialistes*, 2 vols. Paris: Giard et Brière. Cited passages in vol. I (1902), pp. 35–36 and 117–118.

Pareto, Vilfredo (1906). *Manuale di economia politica*. Milano: Società Editrice Libraria.

Pareto, Vilfredo (1916). *Trattato di sociologia generale*. Firenze: Barbèra. English: *The Mind and Society*, trans. Andrew Bongiorno and Arthur Livingston, ed. Livingston. New York: Harcourt, Brace, 1935, vol. III, §2053, p. 1430.

Penn World Table. *Penn World Table*, version 11.0. Groningen Growth and Development Centre. Annual hours and trade series accessed through Our World in Data.

Ricardo, David (1821). *On the Principles of Political Economy and Taxation*, 3rd ed., ch. I, sections I and IV.

Samuelson, Paul A. (1971). "Understanding the Marxian Notion of Exploitation: A Summary of the So-Called Transformation Problem Between Marxian Values and Competitive Prices." *Journal of Economic Literature* 9(2), 399–431. Cited through Baumol, William J. (1974), "The Transformation of Values: What Marx 'Really' Meant (An Interpretation)," *Journal of Economic Literature* 12(1), 51–62, p. 52; and Duménil and Foley (2008).

References (18/24)

Smith, Adam (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*, book I, ch. VI.

Steedman, Ian (1977). *Marx after Sraffa*. London: New Left Books.

Thomas, Ryland and Nicholas Dimsdale (2017). *A Millennium of Macroeconomic Data for the UK*. Bank of England, version 3.1. Real earnings, national income and labor-share series.

Thorp, Willard L. and Hildegarde E. Thorp (1926). "The Annals of Russia," in *Business Annals*. New York: NBER, 234–244.

References (19/24)

Tocqueville, Alexis de (1835). “Mémoire sur le paupérisme.” English: “Memoir on Pauperism,” trans. Seymour Drescher, *The Public Interest* (Winter 1983), 102–120; quoted passage, pp. 104–105.

Voth, Hans-Joachim (2001). “The Longest Years: New Estimates of Labor Input in England, 1760–1830,” *Journal of Economic History* 61(4), 1065–1082.

West, Edwin G. (1983). “Marx’s Hypotheses on the Length of the Working Day,” *Journal of Political Economy* 91(2), 266–281.

Williamson, Jeffrey G. (1990). *Coping with City Growth during the British Industrial Revolution*. Cambridge University Press.

References (20/24)

World Bank. *World Development Indicators*, indicator NE.TRD.GNFS.ZS: trade in goods and services as a share of GDP. Accessed through Our World in Data.

Wright, Gavin (2006). *Slavery and American Economic Development*. Louisiana State University Press, p. 60; wealth figures cited through MeasuringWorth.

Zwiedineck-Südenhorst, Otto von (1955). “Böhm-Bawerk (Böhm von Bawerk), Eugen Ritter von,” *Neue Deutsche Biographie* 2, 385–386.

References (21/24)

Baysal, Ercan (2010). *Injustice*. Cartoon, reposted by the artist in 2022. Reproduced for classroom analysis.

Clarke, Christopher (2026). *Where billionaires are common, poverty is rare*, chart posted 2 November. Forbes World's Billionaires 2026 and World Bank Poverty and Inequality Platform; latest surveys, 2021 PPP.

United States working-hours legislation. Library of Congress, Adamson Act (1916); U.S. Department of Labor, FLSA history and overtime rules.

French working-hours legislation. Assemblée nationale, 1919 and 1936 laws; Raffin, Charles and Hatice Yildiz (2019), *Depuis 1975, le temps de travail annuel a baissé de 350 heures*, Insee.

References (22/24)

Crafts, Nicholas (2020). “Slow Real Wage Growth during the Industrial Revolution: Productivity Paradox or Pro-Rich Growth?” Warwick Economics Research Papers 1268, Tables 2–4.

Geloso, Vincent (forthcoming 2027). *The Great Enrichment: A New History of American Growth and Inequality, 1865–1945*. University of Chicago Press. Author’s online preview, ch. 2, accessed 4 November 2026.

Strain, Michael R. (2019). “The Link Between Wages and Productivity Is Strong.” In Melissa S. Kearney and Amy Ganz (eds.), *Expanding Economic Opportunity for More Americans*. Aspen Institute, 168–179.

References (23/24)

Economic Policy Institute (2026). “The Productivity–Pay Gap.” Quarterly data through 2026 Q2, release v2026.9.14. Data and methodology accessed 4 November 2026.

U.S. Bureau of Labor Statistics (2026). *Productivity and Costs*, nonfarm business sector, via FRED: OPHNFB, COMPNFB, COMPRNFB and IPDNBS. Quarterly data through 2026 Q2, accessed 4 November 2026. See also Brill et al. (2017), “Understanding the Labor Productivity and Compensation Gap.”

Catherine, Sylvain, Max Miller, James D. Paron and Natasha Sarin (2026). *Who Gains When Interest Rates Fall?* Working paper, 20 March, pp. 15–16, Fig. 3B. Wealth shares from Piketty, Saez and Zucman (2018); estimated 10-year real forward rates.

References (24/24)

de Rugy, Veronique (2022). “What Productivity–Pay Gap?” Econlib, 9 December.

Lawrence, Robert Z. (2016). “Does Productivity Still Determine Worker Compensation? Domestic and International Evidence.” In Michael R. Strain (ed.), *The US Labor Market: Questions and Challenges for Public Policy*. AEI Press. Charts reproduced in Strain (2019), pp. 175–176.

Stansbury, Anna M. and Lawrence H. Summers (2017). “Productivity and Pay: Is the Link Broken?” NBER Working Paper 24165, December.